



Dr. Hitaishi V

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Education

2021 - 2025

Ph.D

Nonlinear optics
& Photonics

VIT-AP University, AP, India

2021-2024

Gate Qualified

82 percentile

2018 - 2020

Master of Science (Physics)

JSS College of Arts, Commerce & Science, Ooty Road, Mysuru

2015 - 2018

Bachelor of Science

Dayananada Sagar college of arts science and commerce, Bengaluru

2009

Diploma course

Basic Computer application and Hardware installation and trouble-shooting with assemble

Hello, I am Hitaishi V,

I am a self driven researcher, passionate about finding novel solutions for complex problems in field of photonics and lasers. My high motivation towards the field lead to some research articles. I am seeking postdoc as a platform to achieve my research goals, building research communities and improving the institutes reputation are my motto.

Experience

I have a strong analytical skills and ideas of principles of FDTD (Finite-Difference Time-Domain) and FEM (Finite Element method) for simulations and to investigate the propagation of light with the waveguide designs. I have modeled and executed simulations on waveguide, resulting in the listed publications. I have mentored lab classes for the students to teach experiments mentioned in syllabus. I am skilled in many software and coding languages like Ansys Lumerical, COMSOL Multiphysics, Origin, Python, Rust, MATLAB etc. Eager to apply these skills to further explore exploration and research in the field of photonics and lasers.

I have been learning photonics based softwares like GDS Factory, Femwell and MEEP for better understanding and reliable open-source alternatives.

Skills

Python	★★★★★
MATLAB	★★★★★
Ansys Lumerical	★★★★
Linux	★★★★
Latex	★★★★
Typst	★★★★
COMSOL	★★★
Photoshop	★★★
Illustrator	★★
Rust	★★

About Me

Father name
Virupaksha

Gender
Male

Marital status
Married

Date of birth
13/01/1998

Nationality
Indian

Languages known
English, Hindi, Kannada

Permanent address
Sri Nidhi Enterprises,
Gandhi Nagar, Turuvekere,
Karnataka - 572227

Publications

- [1] H. V., N. Ashok, S. K. Reddy, and S. Pidishety, "Supercontinuum Generation: A Comprehensive Review of Experimental and Material Advances," *Applied Materials Today*, vol. 45, p. 102804, Aug. 2025, doi: 10.1016/j.apmt.2025.102804.
- [2] J. Kulanthaivel, V. Hitaishi, and N. Ashok, "A Tamm-Fano Resonance Glucose Sensor Based on Cu and Distributed Bragg Reflector Plasmonic Coupling Interface in the near-Infrared Regime," *Optical and Quantum Electronics*, vol. 56, no. 9, p. 1538, Sep. 2024, doi: 10.1007/s11082-024-07396-2.
- [3] V. Hitaishi, J. Kulanthaivel, and N. Ashok, "Effects of Waveguide Dimensions on Split-Core Optical Waveguide for Dispersion and Supercontinuum Applications," *Journal of Computational Electronics*, vol. 24, no. 3, p. 71, Jun. 2025, doi: 10.1007/s10825-025-02311-8.
- [4] V. Hitaishi, J. Kulanthaivel, and N. Ashok, "Study of Supercontinuum Generation in Etch-Coated Tellurium Based Chalcogenide Waveguide," *Optical and Quantum Electronics*, vol. 56, no. 11, p. 1814, Nov. 2024, doi: 10.1007/s11082-024-07743-3.
- [5] V. Hitaishi and A. Nandam, "Mid-IR Supercontinuum Generation Using a Slab with D-shaped/Semi-Ellipse Core Waveguide at 4 μ m," *Optical Materials*, vol. 147, p. 114668, Jan. 2024, doi: 10.1016/j.optmat.2023.114668.
- [6] V. Hitaishi, J. K., and N. Ashok, "Inverted Rib-Core Chalcogenide Waveguide for Supercontinuum Generation," in *2023 IEEE Workshop on Recent Advances in Photonics (WRAP)*, Prayagraj, India: IEEE, Dec. 2023, pp. 1–3. doi: 10.1109/WRAP59682.2023.10712858.
- [7] V. Hitaishi, K. Jayakrishnan, and N. Ashok, "Design and Analysis of Chalcogenide GeAsSe Waveguide for Dispersion Properties," *Advanced Nanomaterials and Their Applications*, vol. 22. Springer Nature Singapore, Singapore, pp. 87–96, 2023. doi: 10.1007/978-981-99-1616-0_9.
- [8] V. Hitaishi and N. Ashok, "Broadband Supercontinuum Generation Using Dispersion Engineered As₂Se₃-GeAsSe-GeAsS Waveguide at 6 μ m," *IEEE Access*, vol. 11, pp. 12294–12302, 2023, doi: 10.1109/ACCESS.2023.3242692.
- [9] J. Kulanthaivel, V. Hitaishi, and N. Ashok, "Slot Waveguide Microring Resonator Based on Silicon Nitride for Refractive Index Sensing," in *2022 IEEE International Conference on Nanoelectronics, Nanophotonics, Nanomaterials, Nanobioscience & Nanotechnology (5NANO)*, Kottayam, India: IEEE, Apr. 2022, pp. 1–3. doi: 10.1109/5NANO53044.2022.9828972.

References

Dr. Nandam Ashok

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Sunil Kumar

Assistant professor,
Department of Physics,
JSSACS, Ooty road, Mysuru.
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Declaration

I hereby declare that the information furnished above is true to the best of my knowledge

– **Hitaishi V**

Achievements

- ▶ Awarded Appreciation for academic excellence event held on SSLC by ViPraSS Foundation. Dated 16 June 2013 and in 01st September 2013 by Shree Shankari bank, Tumkur
- ▶ Appreciated for the efforts towards serving the society from Youth for Seva on 5th march 2017
- ▶ Won First Place in the Annual college Chess tournament 2018-19 organized by JSS college of arts commerce and science, Mysuru
- ▶ Won Senior 1st Place in District level chess tournament held on 29 June 2014

Presentations

- ▶ Presented poster on “Inverted Rib-core Chalcogenide Wave- guide For Supercontinuum Generation” at WRAP-2023 at IIIT Allahabad during 7th-9th dec 2023.
- ▶ Presented paper on “Design and Analysis of Chalcogenide GeAsSe Waveguide for Dispersion Properties” at the ICANA 2022 during 16th -18th nov 2022.
- ▶ Topic presentation on “Swachh Bharath” in conference on Municipal solid waste management organized on 23 rd and 24th Feb 2016 at IISc, Bengaluru.
- ▶ Presented paper on “Self-sustainable Wards from waste” in an event organized by University of Commons

Participation

- ▶ Participated in One day international Conference on “Science and technology for management of emerging environmental issues” held in 7th January 2017 at Jnana Jyothi Auditorium Bengaluru. Organized by Department of Environmental science, Bangalore University
- ▶ Participated in Two day Workshop on “Indian Mathematics, their works and “Astronomy” held on 17th and 18th February 2017 organized by Department of Mathematics, Global Academy of Technology with ISRO.
- ▶ Participated in UGC Sponsored State level seminar on “Role of Nuclear science in energy production” held on March 1 st & 2nd 2019 organized by JSS college of arts, commerce, and science.
- ▶ Participated in two-day UGC sponsored National seminar on “Environmental Degradation and its effects on substance of Life” organized by Department of Botany, Government First grade college, vijayanagara, Bengaluru. In association with Karnataka State pollution Control board held on 4th and 5th of march 2016.
- ▶ Participated in pre-university state level chess tournament 2013-14 at Chamarajanagar. Organized by Department of Karnataka pre university Education, malleshwaram Bengaluru.